

# Olympiad Test Paper — Mathematics (Class 7) — Paper 4

## Chapter 12: Another Peek Beyond the Point

### Paper 4 — (progressively harder)

|                        |                                      |
|------------------------|--------------------------------------|
| <b>Subject</b>         | Mathematics (NCERT Class 7)          |
| <b>Chapter</b>         | 12 — Another Peek Beyond the Point   |
| <b>Total Questions</b> | 60 (Multiple Choice, single correct) |
| <b>Maximum Marks</b>   | 240                                  |
| <b>Suggested Time</b>  | 90 minutes                           |

**Marking scheme:** +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

**Instructions:** Each question has four options (A), (B), (C), (D); exactly one is correct. The deadliest slips are miscounting decimal places in a product, scaling only one number when dividing by a decimal, and confusing a place-value shift ( $\times 100$ ) with appending zeros. Do not write on this paper — mark answers on your answer sheet.

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1. Compute  $0.24 \times 0.35$ .

- (A) 0.84 (B) 0.0084  
(C) 8.4 (D) 0.084

2. Compute  $0.0035 \times 100000$ .

- (A) 35 (B) 350  
(C) 3500 (D) 3.5

3. Estimate first:  $51.8 \times 0.49$  is closest to which whole number?

- (A) 250 (B) 2.5  
(C) 25 (D) 260

4. Compute  $6.3 \div 0.018$ .

- (A) 35 (B) 3500  
(C) 3.5 (D) 350

5. Write 0.625 as a fraction in lowest terms.

- (A) 625/100 (B) 5/8  
(C) 5/80 (D) 63/100

6. In the number 70.408, what is the place value of the digit 4?

(A) 4 hundredths (0.04)

(B) 4 thousandths (0.004)

(C) 4 tenths (0.4)

(D) 4 ones (4)

7. Compute  $1.25 \times 0.48$ .

(A) 6

(B) 0.06

(C) 0.6

(D) 60

8. Compute  $(0.04)^2$ .

(A) 0.016

(B) 0.16

(C) 0.0016

(D) 0.00016

9. Compute  $4.7 \times 1000 \div 100$ .

(A) 47

(B) 4.7

(C) 470

(D) 4700

10. A number multiplied by 0.08 gives 0.24. The number is:

(A) 0.3

(B) 30

(C) 0.03

(D) 3

11. Compute  $0.084 \div 0.0012$ .

(A) 7

(B) 700

(C) 0.7

(D) 70

12. Estimate first, then give the exact value of  $0.392 \div 0.014$ .

(A) 2.8

(B) 280

(C) 0.28

(D) 28

13. How many decimal places will the product  $0.015 \times 0.24$  have?

(A) 5

(B) 4

(C) 6

(D) 3

14. Which fraction equals 0.024?

(A)  $3/125$

(B)  $24/100$

(C)  $3/25$

(D)  $3/1250$

15. Compute  $(3.6 \div 0.9) \times 0.05$ .

(A) 2

(B) 0.02

(C) 0.2

(D) 20

16. By what single power of ten must 0.0006 be multiplied to give 6?

(A) 1000

(B) 100000

(C) 100

(D) 10000

17. Petrol costs ₹104.50 per litre. The cost of 0.4 litre is:

(A) ₹41.80

(B) ₹418

(C) ₹4.18

(D) ₹4180

**18.** A product was computed as 9.6. Which of the following correctly produces 9.6?

(A)  $24 \times 4$

(B)  $2.4 \times 4$

(C)  $0.24 \times 4$

(D)  $2.4 \times 40$

**19.** Compute  $12 \div 0.024$ .

(A) 50

(B) 5000

(C) 500

(D) 5

**20.** Compute  $(1.2 + 0.8) \div 0.05$ .

(A) 40

(B) 4

(C) 400

(D) 0.4

**21.** Estimate first, then give the exact value of  $29.6 \times 0.25$ .

(A) 7.4

(B) 74

(C) 0.74

(D) 740

**22.** Compute  $0.045 \times 0.6$ .

(A) 0.27

(B) 0.027

(C) 0.0027

(D) 2.7

**23.** Compute  $(0.2)^3 \times 100$ .

(A) 0.8

(B) 8

(C) 0.08

(D) 80

**24.** In 6.49, by how much does the value change if the digit 4 is replaced by 9?

(A) increases by 5

(B) increases by 0.05

(C) increases by 0.45

(D) increases by 0.5

**25.** Rice costs ₹52.50 per kg. A shopper buys 1.6 kg and pays with a ₹100 note. The change received is:

(A) ₹84

(B) ₹16

(C) ₹1.60

(D) ₹160

**26.** Compute  $0.6 \times 1000 \div 100 \times 10$ .

(A) 6

(B) 600

(C) 60

(D) 0.6

**27.** The product of two numbers is 0.0036. If one of them is 0.09, the other is:

(A) 0.4

(B) 0.004

(C) 0.04

(D) 4

**28.** Compute  $18 \div 0.0225$ .

(A) 80

(B) 8000

(C) 8

(D) 800

**29.** Compute  $2.5 \times 0.8 + 1.5 \times 0.8$ .

- (A) 32 (B) 3.2  
(C) 0.32 (D) 2.4

**30.** A number written as 5.736 is multiplied by 100. In the result, which digit is in the ones (units) place?

- (A) 5 (B) 3  
(C) 7 (D) 6

**31.** Compute  $0.36 \times 0.36$ .

- (A) 1.296 (B) 0.01296  
(C) 12.96 (D) 0.1296

**32.** A plank 3.6 m long has three pieces of 0.95 m, 0.8 m and 0.65 m cut off. The remaining length is:

- (A) 1.2 m (B) 2.4 m  
(C) 0.2 m (D) 1.8 m

**33.** Compute  $12.5 \times 1000 \div 10000$ .

- (A) 12.5 (B) 0.125  
(C) 125 (D) 1.25

**34.** Express  $13/8$  as a decimal.

- (A) 1.375 (B) 1.625  
(C) 0.625 (D) 1.0625

**35.** Estimate first:  $0.51 \div 0.0098$  is closest to:

- (A) 50 (B) 5  
(C) 500 (D) 0.5

**36.** A bag of sugar weighs 30.6 kg. It is repacked equally into 0.85 kg packets. The number of packets is:

- (A) 36 (B) 3.6  
(C) 360 (D) 26.01

**37.** Honey costs ₹3.60 per millilitre. A jar holds 0.45 litre (= 450 mL). The jar's contents are worth:

- (A) ₹162 (B) ₹16200  
(C) ₹1620 (D) ₹16.20

**38.** Compute  $0.072 \times 10000 \div 1000$ .

- (A) 7.2 (B) 0.72  
(C) 0.072 (D) 72

**39.** Compute  $(0.05)^3$ .

- (A) 0.00125  
(C) 0.000125

- (B) 0.0125  
(D) 0.0000125

**40.** Compute  $7.2 \div (0.6 \times 0.3)$ .

- (A) 4  
(C) 400

- (B) 40  
(D) 0.4

**41.** Compute  $4.8 \div 0.0016$ .

- (A) 3000  
(C) 30000

- (B) 300  
(D) 30

**42.** A tank holds 9.6 litres. A jug holds 0.35 litre. How many full jugs can be filled, and how much is left?

- (A) 27 jugs, 0 litres left  
(C) 27 jugs, 0.15 litre left

- (B) 2.7 jugs  
(D) 28 jugs, 0.15 litre left

**43.** Compute  $0.018 \div 0.4$ .

- (A) 0.45  
(C) 4.5

- (B) 0.0045  
(D) 0.045

**44.** Compute  $0.125 \times 0.64$ .

- (A) 0.8  
(C) 0.08

- (B) 0.008  
(D) 8

**45.** Compute  $5.6 \times 1000 - 5.6 \times 100$ .

- (A) 504  
(C) 50400

- (B) 5040  
(D) 5.04

**46.** Which decimal equals  $7/250$ ?

- (A) 0.28  
(C) 2.8

- (B) 0.0028  
(D) 0.028

**47.** If 1.5 metres of ribbon cost ₹67.50, the cost of 4 metres at the same rate is:

- (A) ₹18  
(C) ₹1800

- (B) ₹180  
(D) ₹270

**48.** Estimate first:  $0.0392 \div 0.0049$  is closest to:

- (A) 0.8  
(C) 8

- (B) 80  
(D) 0.08

**49.** A nail is 0.025 m long. Laid end to end, how many nails make a line 1.5 m long?

- (A) 6  
(C) 0.06

- (B) 600  
(D) 60

**50.** In the number 2.0506, what is the place value of the digit 5?

(A) 5 thousandths (0.005)

(B) 5 tenths (0.5)

(C) 5 ten-thousandths (0.0005)

(D) 5 hundredths (0.05)

**51.** Compute  $(0.4 + 0.6) \div 0.008$ .

(A) 12.5

(B) 125

(C) 1250

(D) 1.25

**52.** Compute  $8.5 \times 0.5 \div 0.05$ .

(A) 85

(B) 8.5

(C) 850

(D) 0.85

**53.**  $(0.06)^2$  is how many times  $(0.06 \div 100)$ ?

(A) 60

(B) 600

(C) 6

(D) 0.6

**54.** Tickets cost ₹62.50 each. A group spends ₹500 on tickets, then shares ₹60 of snacks equally among the ticket-holders. Each person's snack share is:

(A) ₹0.75

(B) ₹75

(C) ₹6.25

(D) ₹7.50

**55.** A ribbon 2.4 m long is cut into 8 equal pieces, and each piece is then cut into 3 equal parts. The length of one final part is:

(A) 0.3 m

(B) 0.01 m

(C) 0.1 m

(D) 0.8 m

**56.** Compute  $(1.2 \times 0.5) \div (0.4 \times 0.3)$ .

(A) 0.5

(B) 5

(C) 50

(D) 0.05

**57.** Express  $9/40$  as a decimal.

(A) 0.225

(B) 0.0225

(C) 0.45

(D) 2.25

**58.** A block of metal weighs 1.44 kg. It is cast into beads each weighing 0.018 kg. The number of beads is:

(A) 80

(B) 8

(C) 800

(D) 0.0125

**59.** A pipe leaks 0.045 litre per minute. The water lost in 2 hours is:

(A) 5.4 litres

(B) 54 litres

(C) 0.54 litre

(D) 0.09 litre

**60.** A 0.4 kg packet of nuts costs ₹150. The cost of 2.5 kg of the same nuts is:

(A) ₹375

(B) ₹937.50

(C) ₹93.75

(D) ₹600

# Solutions — Mathematics (Class 7), Chapter 12: Another Peek Beyond the Point — Paper 4

## Paper 4 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

## Answer Key

| Q  | Ans | Q  | Ans | Q  | Ans | Q  | Ans | Q  | Ans | Q  | Ans |
|----|-----|----|-----|----|-----|----|-----|----|-----|----|-----|
| 1  | D   | 11 | D   | 21 | A   | 31 | D   | 41 | A   | 51 | B   |
| 2  | B   | 12 | D   | 22 | B   | 32 | A   | 42 | C   | 52 | A   |
| 3  | C   | 13 | A   | 23 | A   | 33 | D   | 43 | D   | 53 | C   |
| 4  | D   | 14 | A   | 24 | D   | 34 | B   | 44 | C   | 54 | D   |
| 5  | B   | 15 | C   | 25 | B   | 35 | A   | 45 | B   | 55 | C   |
| 6  | C   | 16 | D   | 26 | C   | 36 | A   | 46 | D   | 56 | B   |
| 7  | C   | 17 | A   | 27 | C   | 37 | C   | 47 | B   | 57 | A   |
| 8  | C   | 18 | B   | 28 | D   | 38 | B   | 48 | C   | 58 | A   |
| 9  | A   | 19 | C   | 29 | B   | 39 | C   | 49 | D   | 59 | A   |
| 10 | D   | 20 | A   | 30 | B   | 40 | B   | 50 | D   | 60 | B   |

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

## Worked Solutions

**1. (D)**  $24 \times 35 = 840$ . The factors have  $2 + 2 = 4$  decimal places, so the product needs 4 decimal places:  $0.0840 = 0.084$ . Counting only 3 places gives the wrong 0.84.

**2. (B)** Multiplying by 100000 shifts the point 5 places right. Starting from 0.0035, five shifts give 350. Shifting only 4 places gives the wrong 35.

**3. (C)** Round to  $50 \times 0.5 = 25$ . The exact value 25.382 rounds to 25; misplacing the point to 253.82 gives 250.

**4. (D)** Scale both by 1000:  $6300 \div 18 = 350$ . Scaling only the divisor ( $6.3 \div 18 = 0.35$ ) then mishandling gives 35.

**5. (B)**  $0.625 = 625/1000$ . Dividing top and bottom by 125 gives  $5/8$ . Stopping at  $625/100$  forgets there are three decimal places (denominator 1000).

- 6. (C)** The first digit after the point is the tenths place, so 4 stands for 4 tenths = 0.4. Reading it as hundredths confuses the position by one place.
- 7. (C)**  $125 \times 48 = 6000$ . The factors carry  $2 + 2 = 4$  decimal places, so  $0.6000 = 0.6$ . Treating the trailing zeros as extra place value gives 6 or 0.06.
- 8. (C)**  $4 \times 4 = 16$ . Each factor 0.04 has 2 decimal places, so the square needs  $2 + 2 = 4$  places: 0.0016. Counting only 3 places gives 0.016.
- 9. (A)**  $\times 1000$  shifts the point 3 right (4700), then  $\div 100$  shifts it 2 left (47). Net effect is one shift right. Forgetting the division leaves 4700.
- 10. (D)** Divide:  $0.24 \div 0.08$ . Scale both by 100:  $24 \div 8 = 3$ . Treating it as  $0.24 \div 8 = 0.03$  forgets to scale the divisor.
- 11. (D)** Scale both by 10000:  $840 \div 12 = 70$ . Scaling unevenly ( $84 \div 12 = 7$ ) leaves a factor of ten missing.
- 12. (D)** Estimate  $0.4 \div 0.014 \approx 28$ . Exactly: scale both by 1000,  $392 \div 14 = 28$ . Misplacing the point gives 2.8 or 280.
- 13. (A)** 0.015 has 3 places and 0.24 has 2 places, so the product has  $3 + 2 = 5$  decimal places (any trailing zeros are still counted before simplifying).
- 14. (A)**  $0.024 = 24/1000$ . Dividing top and bottom by 8 gives  $3/125$ . Writing  $24/100$  uses the wrong denominator (only two places);  $3/25 = 0.12$ .
- 15. (C)**  $3.6 \div 0.9 = 4$  ( $36 \div 9$ ). Then  $4 \times 0.05 = 0.20 = 0.2$ . Doing  $4 \times 0.5$  by misreading the factor gives 2.
- 16. (D)** Moving the point from 0.0006 to 6.0 is 4 places right, which is  $\times 10000$ . Counting only 3 places gives 1000 ( $\rightarrow 0.6$ , not 6).
- 17. (A)**  $104.50 \times 0.4$ :  $1045 \times 4 = 4180$ , with  $2 + 1 = 3$  decimal places  $\rightarrow 41.80$ . Forgetting the 0.4 has one decimal place gives ₹418.
- 18. (B)**  $2.4 \times 4 = 9.6$  ( $24 \times 4 = 96$ , one decimal place).  $24 \times 4 = 96$  and  $2.4 \times 40 = 96$ , both ten times too big;  $0.24 \times 4 = 0.96$ , ten times too small.
- 19. (C)** Scale both by 1000:  $12000 \div 24 = 500$ . Scaling only the divisor ( $12 \div 24 = 0.5$ ) misses three zeros.
- 20. (A)**  $1.2 + 0.8 = 2.0$ , then  $2 \div 0.05$ : scale both by 100,  $200 \div 5 = 40$ . Dividing  $2 \div 5 = 0.4$  forgets to scale the divisor.
- 21. (A)** Estimate  $30 \times 0.25 = 7.5$ , so the answer is near 7.4. Exactly:  $296 \times 25 = 7400$ , with  $1 + 2 = 3$  places  $\rightarrow 7.400 = 7.4$ .



- 22. (B)**  $45 \times 6 = 270$ . The factors have  $3 + 1 = 4$  decimal places, so  $0.0270 = 0.027$ . Counting only 3 places gives 0.27.
- 23. (A)**  $(0.2)^3 = 0.008$  ( $2 \times 2 \times 2 = 8$ , three decimal places). Then  $0.008 \times 100$  shifts the point 2 right = 0.8. Stopping at one shift gives 0.08.
- 24. (D)** The 4 is in the tenths place worth 0.4; replacing it with 9 makes it 0.9, an increase of 0.5. Reading it as hundredths gives the wrong 0.05.
- 25. (B)** Cost =  $52.50 \times 1.6 = 84.00$  ( $525 \times 16 = 8400$ , three decimal places). Change =  $100 - 84 = 16$ . Confusing cost with change gives ₹84.
- 26. (C)** Left to right:  $0.6 \times 1000 = 600$ ;  $\div 100 = 6$ ;  $\times 10 = 60$ . Dropping the final  $\times 10$  gives 6.
- 27. (C)**  $0.0036 \div 0.09$ : scale both by 100,  $0.36 \div 9 = 0.04$ . Dividing  $0.0036 \div 9 = 0.0004$  forgets to scale the divisor.
- 28. (D)** Scale both by 10000:  $180000 \div 225 = 800$ . Estimating  $18 \div 0.02 \approx 900$  confirms order of magnitude; 80 is ten times too small.
- 29. (B)** Factor out 0.8:  $(2.5 + 1.5) \times 0.8 = 4 \times 0.8 = 3.2$ . Adding only one product (2.0) and adding 1.2 wrongly, or misplacing the point, gives 0.32.
- 30. (B)**  $5.736 \times 100 = 573.6$ . The ones digit is 3 (573.6 has 5 hundreds, 7 tens, 3 ones, 6 tenths). Reading the original ones digit 5 ignores the shift.
- 31. (D)**  $36 \times 36 = 1296$ . The factors have  $2 + 2 = 4$  decimal places, so 0.1296. Counting only 3 places gives 1.296.
- 32. (A)** Cut total =  $0.95 + 0.8 + 0.65 = 2.4$  m. Remaining =  $3.6 - 2.4 = 1.2$  m. Forgetting to subtract from the plank and reporting the cut total gives 2.4.
- 33. (D)**  $\times 1000 \rightarrow 12500$ ;  $\div 10000$  shifts 4 left  $\rightarrow 1.25$ . Net is one shift left from 12.5. Shifting too far gives 0.125.
- 34. (B)**  $13 \div 8 = 1.625$  (8 into 130 is 16 r2  $\rightarrow .6$ ; 8 into 20 is 2 r4  $\rightarrow .02$ ; 8 into 40 is 5  $\rightarrow .005$ ). 1.375 is  $11/8$ , not  $13/8$ .
- 35. (A)** Round to  $0.5 \div 0.01 = 50$ . The exact value 52.04... is near 50; misplacing the point gives 5 or 500.
- 36. (A)**  $30.6 \div 0.85$ : scale both by 100,  $3060 \div 85 = 36$ . Multiplying instead of dividing gives 26.01.
- 37. (C)**  $3.60 \times 450 = 1620$  ( $36 \times 45 = 1620$ , then place the point: 3.60 has 2 places, 450 has 0, product  $1620.00 = 1620$ ). Using 0.45 litre directly forgets the litre  $\rightarrow$  mL conversion.

**38. (B)**  $\times 10000$  shifts 4 right  $\rightarrow 720$ ;  $\div 1000$  shifts 3 left  $\rightarrow 0.72$ . Net one shift right. Stopping after the multiply gives 720; an extra shift gives 7.2.

**39. (C)**  $5 \times 5 \times 5 = 125$ . Each factor 0.05 has 2 decimal places, so the cube needs  $2 + 2 + 2 = 6$  places: 0.000125. Counting only 5 places gives 0.00125.

**40. (B)**  $0.6 \times 0.3 = 0.18$ , then  $7.2 \div 0.18$ : scale both by 100,  $720 \div 18 = 40$ . Dividing by 0.18 as if by 1.8 gives 4.

**41. (A)** Scale both by 10000:  $48000 \div 16 = 3000$ . Scaling unevenly drops a factor of ten, giving 300.

**42. (C)**  $9.6 \div 0.35$ :  $0.35 \times 27 = 9.45$ , and  $0.35 \times 28 = 9.80$  exceeds 9.6, so 27 full jugs; remainder  $9.6 - 9.45 = 0.15$  litre. Ignoring the remainder claims 0 left.

**43. (D)** Scale both by 10:  $0.18 \div 4 = 0.045$ . Dividing  $0.018 \div 4 = 0.0045$  forgets to scale the divisor.

**44. (C)**  $125 \times 64 = 8000$ . The factors have  $3 + 2 = 5$  decimal places, so  $0.08000 = 0.08$ . Counting too few places gives 0.8.

**45. (B)**  $5.6 \times 1000 = 5600$ ;  $5.6 \times 100 = 560$ ; difference  $5600 - 560 = 5040$ . Subtracting before shifting, or one wrong shift, gives 504.

**46. (D)** Multiply top and bottom by 4 to get  $28/1000 = 0.028$ . Using denominator 100 instead ( $28/100 = 0.28$ ) is the common slip.

**47. (B)** Unit rate =  $67.50 \div 1.5 = 45$  per metre. Then  $45 \times 4 = 180$ . Multiplying  $67.50 \times 4 \div 1.5$  the wrong way, or misplacing the point, gives 18.

**48. (C)** Round to  $0.04 \div 0.005 = 8$ . Exactly, scale both by 10000:  $392 \div 49 = 8$ . Misplacing the point gives 0.8 or 80.

**49. (D)**  $1.5 \div 0.025$ : scale both by 1000,  $1500 \div 25 = 60$ . Dividing  $1.5 \div 25 = 0.06$  forgets to scale the divisor.

**50. (D)** After the point the order is tenths(0), hundredths(5), thousandths(0), ten-thousandths(6). So the 5 is in the hundredths place = 0.05.

**51. (B)**  $0.4 + 0.6 = 1.0$ , then  $1 \div 0.008$ : scale both by 1000,  $1000 \div 8 = 125$ . Dividing  $1 \div 8 = 0.125$  then misplacing gives 12.5.

**52. (A)**  $8.5 \times 0.5 = 4.25$ , then  $4.25 \div 0.05$ : scale both by 100,  $425 \div 5 = 85$ . Dividing  $4.25 \div 5 = 0.85$  forgets to scale the divisor.

**53. (C)**  $(0.06)^2 = 0.0036$ .  $(0.06 \div 100) = 0.0006$ . Ratio =  $0.0036 \div 0.0006 = 6$ . Mishandling the place shift gives 60.

**54. (D)** Number of people =  $500 \div 62.50 = 8$ . Snack share =  $60 \div 8 = 7.50$ . Dividing 60 by 80 gives the wrong 0.75.

**55. (C)** Each piece =  $2.4 \div 8 = 0.3$  m; each part =  $0.3 \div 3 = 0.1$  m. Stopping after the first cut gives 0.3.

**56. (B)** Numerator  $1.2 \times 0.5 = 0.6$ ; denominator  $0.4 \times 0.3 = 0.12$ . Then  $0.6 \div 0.12$ : scale both by 100,  $60 \div 12 = 5$ . Dividing  $0.6 \div 12$  gives 0.05.

**57. (A)** Multiply top and bottom by 25 to get  $225/1000 = 0.225$ . Using denominator 10000 (0.0225) shifts the point two places too far.

**58. (A)**  $1.44 \div 0.018$ : scale both by 1000,  $1440 \div 18 = 80$ . Dividing  $1.44 \div 18 = 0.08$  forgets to scale; multiplying gives 0.0259.

**59. (A)** 2 hours = 120 minutes. Loss =  $0.045 \times 120 = 5.4$  litres ( $45 \times 120 = 5400$ , three decimal places  $\rightarrow 5.400$ ). Multiplying by 2 instead of 120 gives 0.09.

**60. (B)** Rate =  $150 \div 0.4 = 375$  per kg. Then  $375 \times 2.5 = 937.50$ . Stopping at the per-kg rate gives 375; misplacing the point gives 93.75.